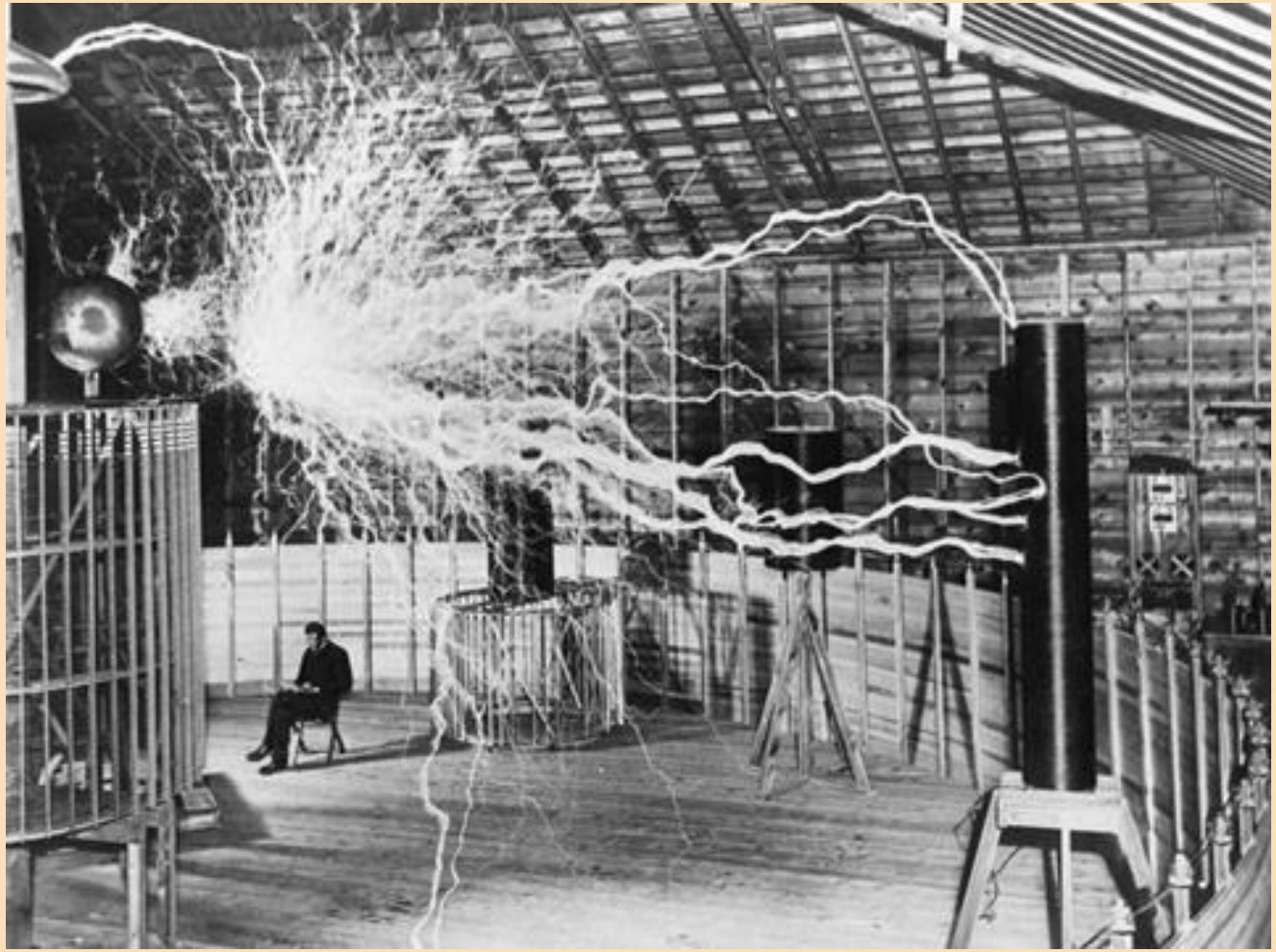
Early Electronics 1880s-1920s

What is going on in the 1870s?

Second Industrial Revolution

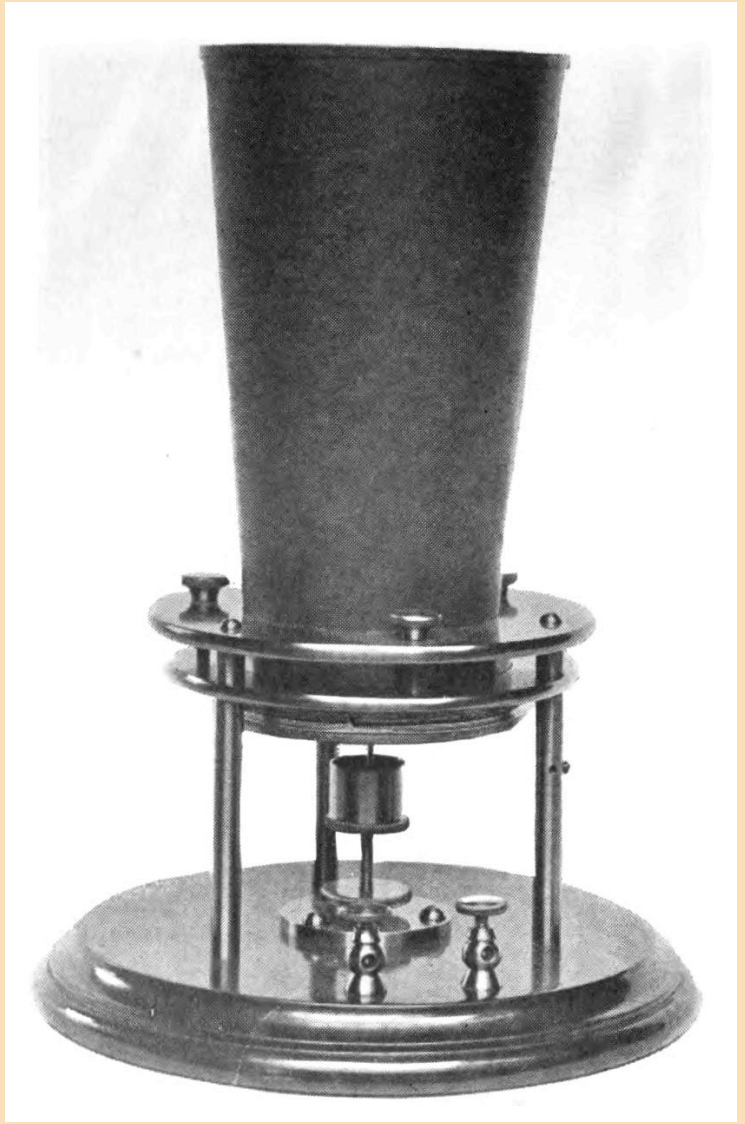
Developments in electricity & communication



What else did we have going into the 1900s?

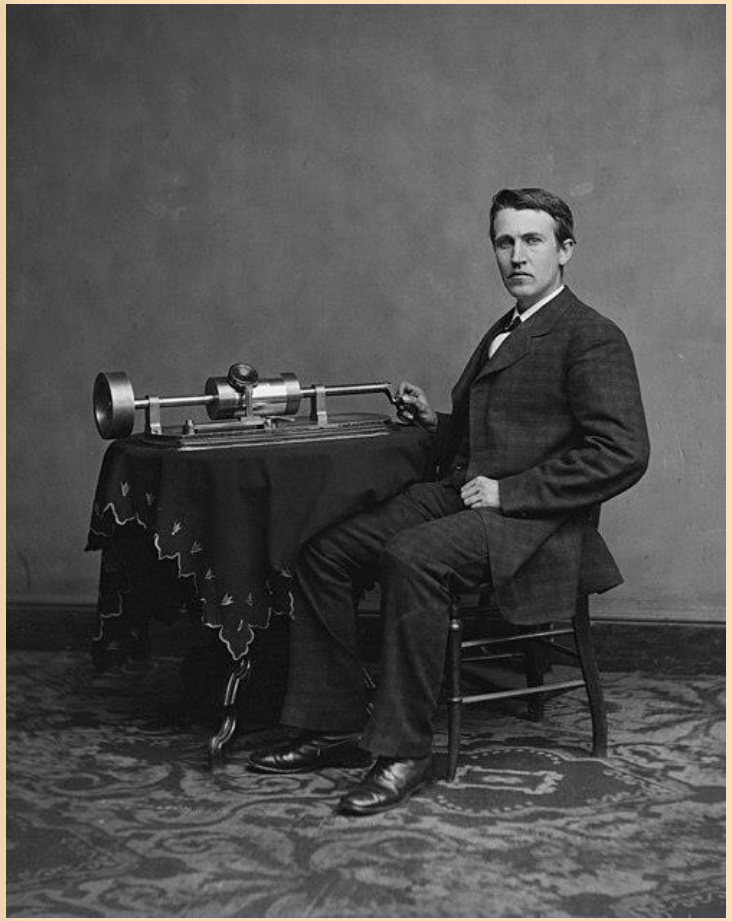
Liquid Transmitter 1876

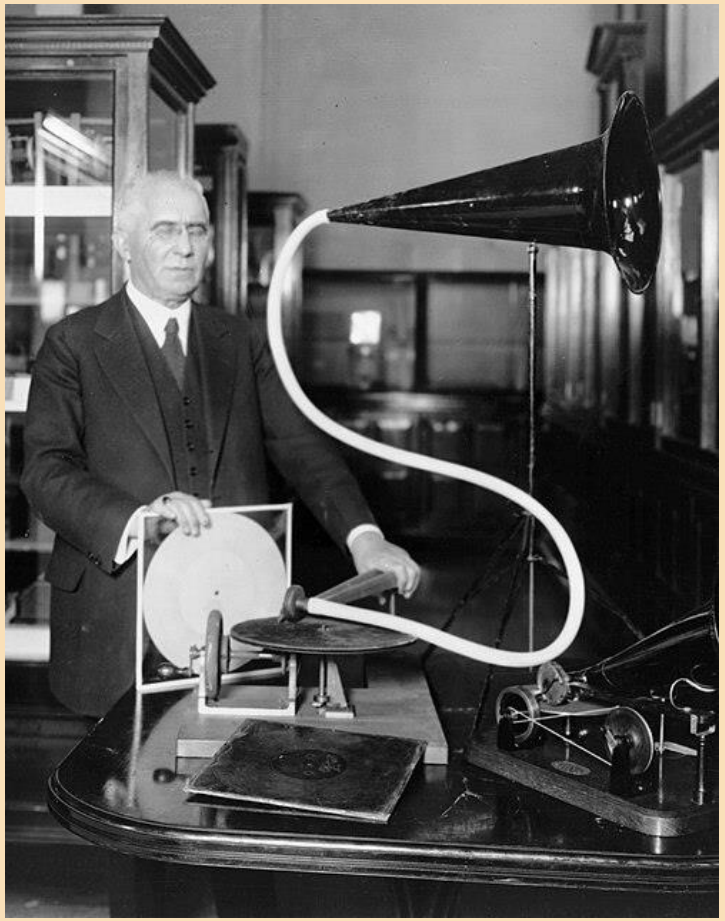
First experimental microphone
Alexander Graham Bell



Phonograph & Gramophone 1877 &1887

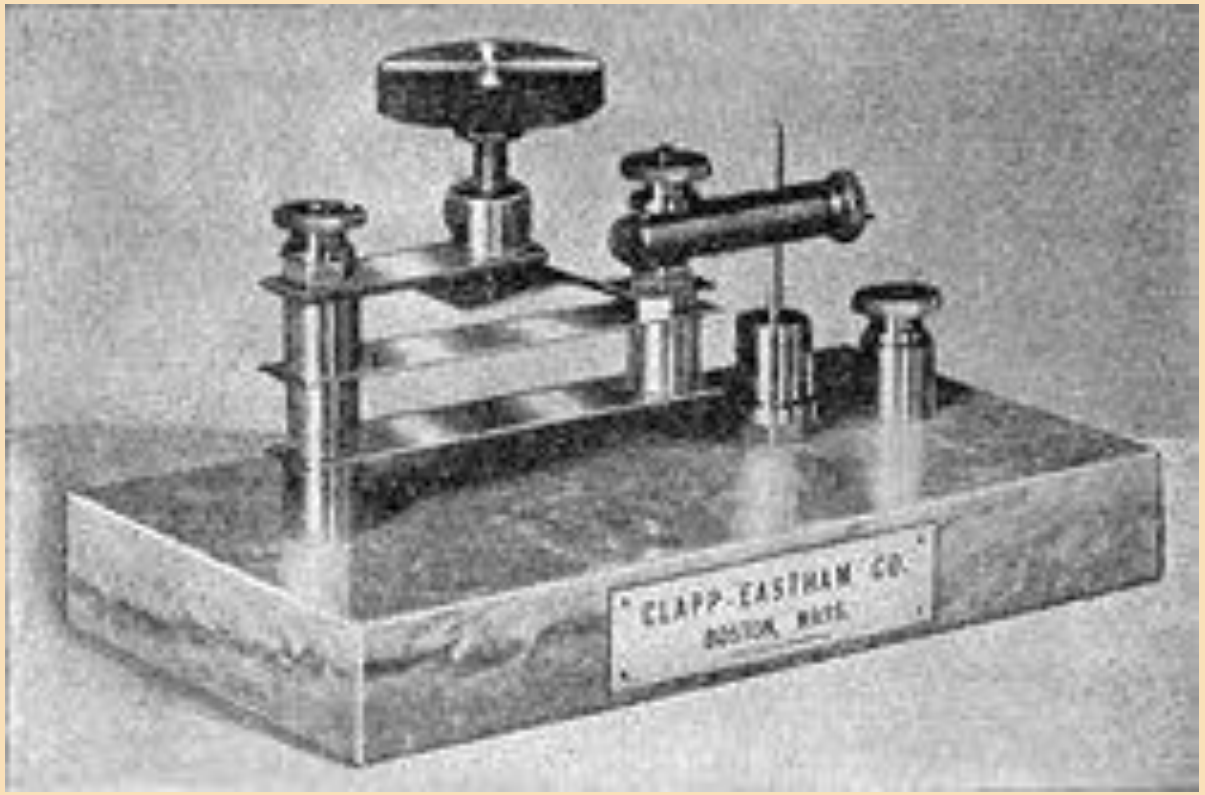
wax cylinders vs. discs
Edison & Emile Berliner





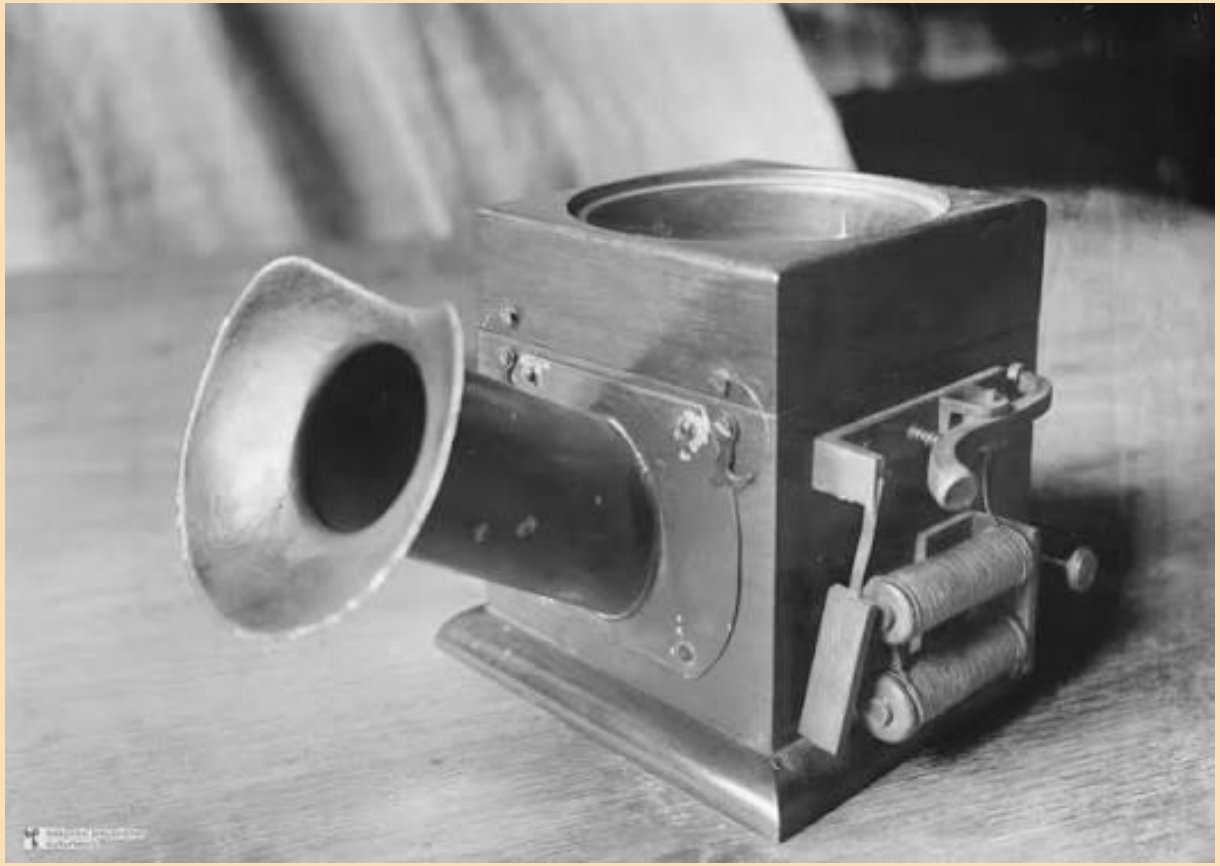
Radio
1895-1920s

Marconi to commercial radio



Transduction

The concept that electricity can be transformed into sound and vice versa



A Transduction Timeline:

1861: German physicist Johann Philipp Reis: "Reis telephone"
Early device that could transmit musical tones and some unintelligible speech

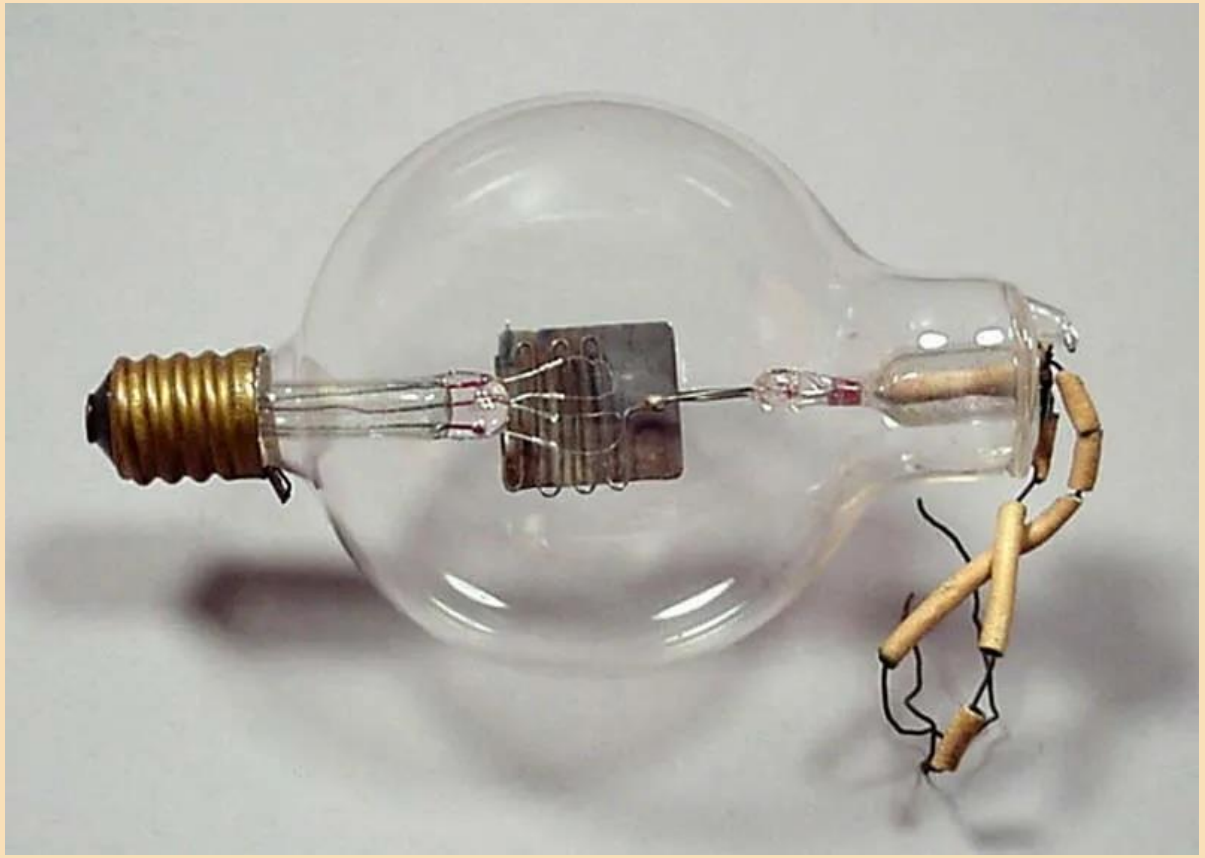
1876: Alexander Graham Bell patented the first commercially successful telephone

1876–1877: Berliner & Edison developed the carbon transmitter/microphone

1878: David Edward Hughes coined the word "microphone"

Lee de Forest's Triode Valve 1906

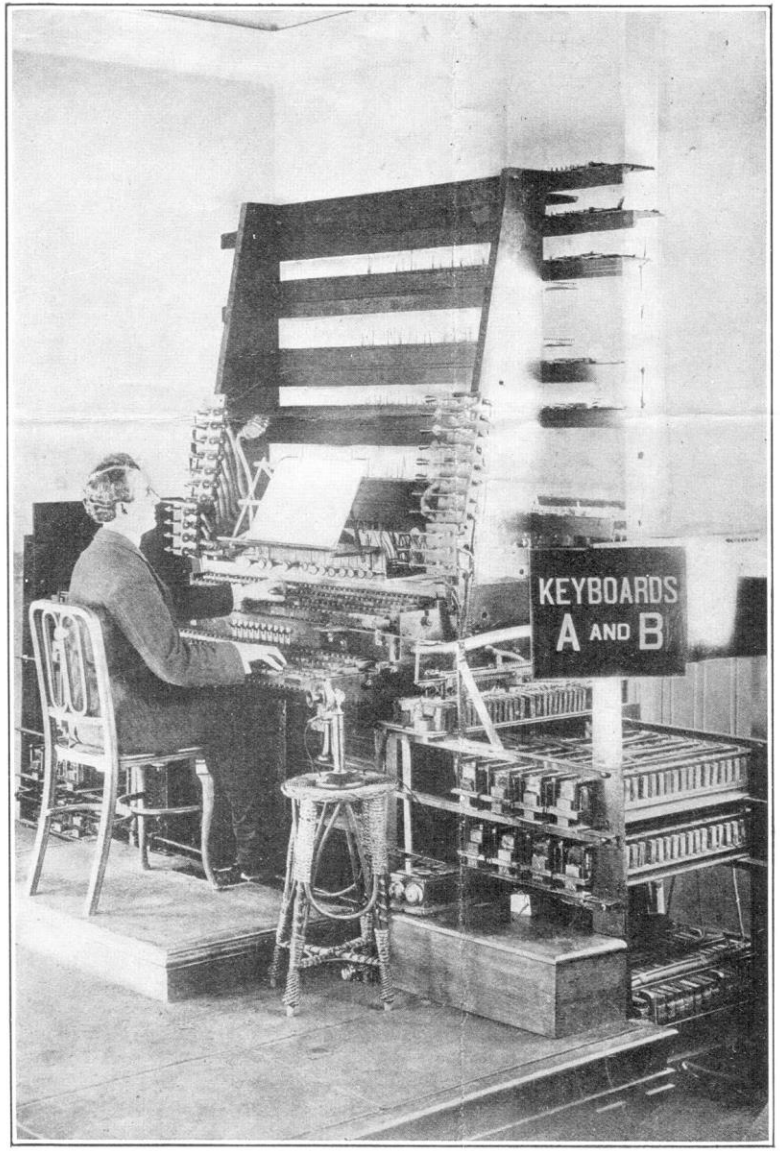
Revolutionized audio amplification



Some early electronic instrument inventions

Telharmonium 1897

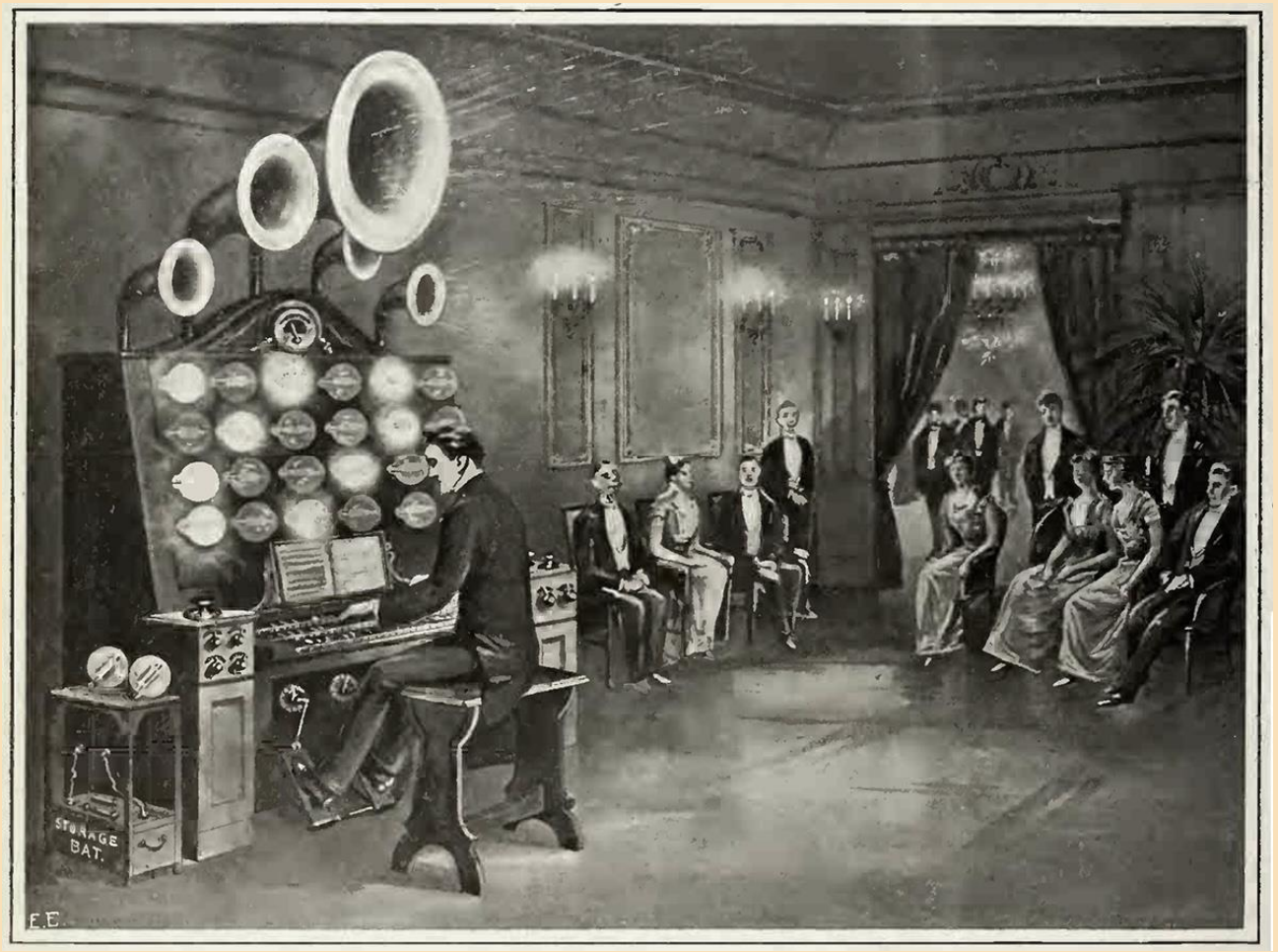
electro-magnetically synthesized anddistributed music over the new telephone networks of victorian America



THE KEYBOARD OF THE TELHARMONIUM

Audion Piano 1915

Audions were what vacuum tubes were initially called by Lee de Forest



Theremin
1922

Heterodyning radio frequencies + body capacitance

Clara Rockmore, performer

https://www.youtube.com/watch?v=_3H5JbkPXpw



Ondes-Martenot 1928

Maurice Martenot, keyboard instrument with slide control

Olivier Messiaen composer

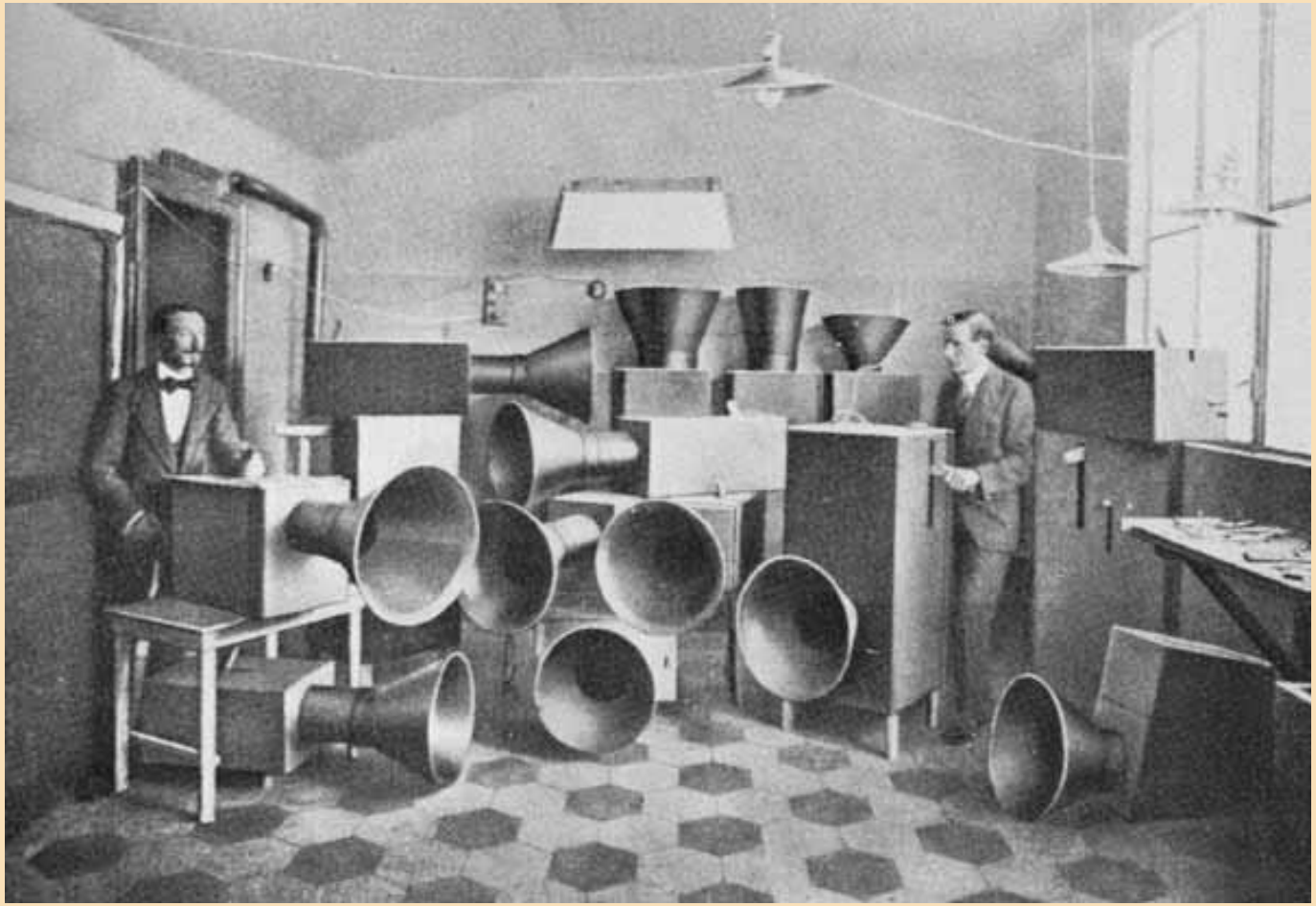
https://youtu.be/6EU0ISo996A?si=V852MuQi-Hqxxh31



Experimenting with acoustics also happened...

Intonarumori 1910

Luigi Russolo
In favor of industrialization

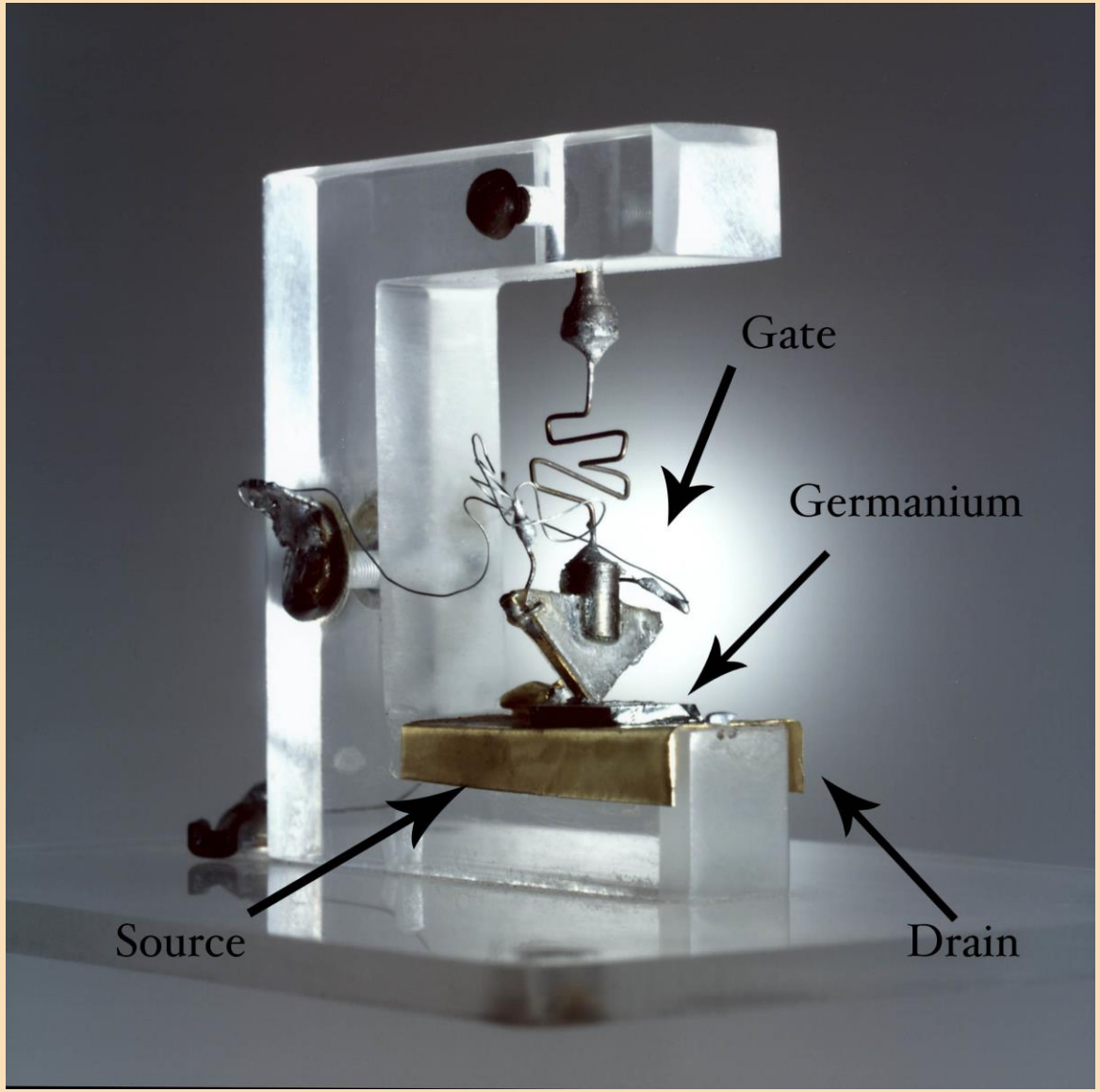


Skipping ahead to the '40s...

First Transistor 1947

Solid state amplification, allowed for smaller electronics and less power/heat

Bardeen, Brattain, Shockley



Replaced tubes in many cases

Tubes were large, hot, required more power, and often burned out





Tubes still have many applications

where overdriving may be a possibility including guitar amps, as solid state doesn't deal with distortion as well, sonically

